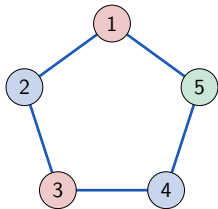


Colorings

Section 1.6

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



Jeremy Quastel

Today's lecture

1. Definitions

2. Bounds on Chromatic Number

3. The Four Color Problem

4. Chromatic Polynomials

A Scheduling Problem

Question

Several exams must be scheduled. Two exams cannot share a time slot when a student is taking both. How few time slots are needed?

Make a graph whose vertices are exams and whose edges represent conflicts.

A time slot will become a color.

Definition: Proper Vertex Coloring

Proper coloring

An assignment of colors to vertices such that adjacent vertices receive different colors.

k -colorable

A graph is k -colorable if it has a proper coloring using at most k colors.

Chromatic number

The least such k is denoted $\chi(G)$.

Examples: Complete Graphs and Paths

Example

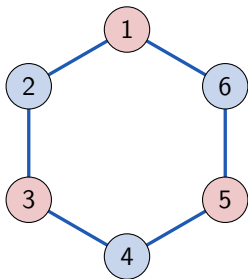
$\chi(K_n) = n$: every two vertices are adjacent.

Example

For $n \geq 2$, $\chi(P_n) = 2$: alternate two colors along the path. At least two are needed because there is an edge.

A nonempty edgeless graph has chromatic number 1.

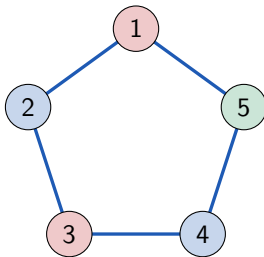
Coloring Cycles



Question

Why do alternating colors work on C_6 ? What goes wrong on an odd cycle?

Odd Cycles Need a Third Color



Cycle chromatic numbers

For $n \geq 3$,

$$\chi(C_n) = \begin{cases} 2, & n \text{ even,} \\ 3, & n \text{ odd.} \end{cases}$$

Theorem: Two Colors and Bipartite Graphs

Equivalent conditions

A graph is 2-colorable exactly when it is bipartite. This is also equivalent to having no odd cycle.

Proof

The two color classes are independent sets, so they form a bipartition. Conversely, color the two parts differently.

A cycle in a bipartite graph alternates between the two parts and is therefore even.

Proof Continued: No Odd Cycle Gives Two Colors

Proof

Work in one connected component and choose a root r . Color a vertex by the parity of its distance from r .

Suppose an edge joined two vertices of the same parity. Their root paths, after removing a common initial portion, together with that edge contain an odd cycle.

Thus every edge joins opposite colors. Repeat in each component. □

Color Classes and Independent Sets

Color class

All vertices assigned one particular color. Every color class is independent.

A k -coloring partitions $V(G)$ into at most k independent sets.

Equivalently, coloring asks how few independent sets are needed to cover the vertices.

Definition: Clique Number

Clique

A set of pairwise adjacent vertices.

Clique number

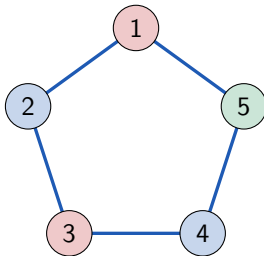
The maximum size of a clique is $\omega(G)$.

Lower bound

$$\chi(G) \geq \omega(G).$$

Every vertex of a clique must have a different color.

A Clique Bound Need Not Be Exact



For C_5 , the largest clique has size 2, but $\chi(C_5) = 3$.

A graph can require more colors than any one clique does.

A Lower Bound from Independence Number

Independence bound

For a graph with n vertices,

$$\chi(G) \geq \left\lceil \frac{n}{\alpha(G)} \right\rceil.$$

Proof

Each color class has at most $\alpha(G)$ vertices. With k classes, $n \leq k\alpha(G)$. Rearrange and use integrality. □

Greedy Coloring

Greedy algorithm

Order the vertices $v_1, \dots, v_n$. At step i , give v_i the smallest positive integer not already used on its colored neighbors.

The output is always a proper coloring.

The number of colors can depend on the ordering.

Greedy Coloring: A Bad Order

Let P_4 have edges ab, bc, cd . Use the order a, d, b, c .

Step	Vertex	Assigned color
1	a	1
2	d	1
3	b	2
4	c	3

The graph needs only two colors, but this ordering makes greedy use three.

Theorem: The Maximum-Degree Bound

Greedy bound

$$\chi(G) \leq \Delta(G) + 1.$$

Proof

When a vertex is colored, at most $\Delta(G)$ of its neighbors have already been colored. They forbid at most $\Delta(G)$ colors, leaving one of the first $\Delta(G) + 1$ colors available. □

A Better Bound from a Good Ordering

Ordering bound

If every vertex has at most d earlier neighbors in an ordering, greedy coloring uses at most $d + 1$ colors.

One way to find such an ordering is to repeatedly delete a vertex of degree at most d , then reverse the deletion order.

The degree condition must hold in every remaining subgraph.

Brooks' Theorem

Brooks

If G is connected and is neither a complete graph nor an odd cycle, then

$$\chi(G) \leq \Delta(G).$$

Complete graphs and odd cycles show why exceptions are needed.

This is a stronger theorem than the greedy bound; we state it without its full proof.

Definition: Critical Graph

k -critical

A graph G with $\chi(G) = k$ such that every proper subgraph has chromatic number less than k .

Every graph of chromatic number k contains a k -critical subgraph: keep deleting vertices or edges whenever doing so preserves the chromatic number.

Critical Graphs Have Large Minimum Degree

Critical-degree bound

A k -critical graph satisfies $\delta(G) \geq k - 1$.

Proof

If some vertex v had degree at most $k - 2$, color $G - v$ with at most $k - 1$ colors. At most $k - 2$ colors appear on the neighbors of v , so a color remains for v .

This produces a $(k - 1)$ -coloring of G , a contradiction. □

Combining Upper and Lower Bounds

Example

Suppose G has 12 vertices, $\alpha(G) = 4$, and $\Delta(G) = 3$. Then

$$3 \leq \chi(G) \leq 4.$$

If G is connected and is not K_4 , Brooks' theorem improves the upper bound to 3, unless it is an odd cycle. Here its order rules out that exception. Thus $\chi(G) = 3$.

Map Coloring and Planar Graphs

Represent each country by a vertex. Join two vertices when the countries share a boundary segment.

Meeting only at a point does not require different colors.

In the usual setting of connected countries, map coloring becomes a planar vertex-coloring problem.

Theorem: Six Colors Suffice

Six Color Theorem

Every simple planar graph is 6-colorable.

Proof

Induct on the number of vertices. A planar graph has a vertex v of degree at most five. Color $G - v$ with six colors by induction. Its neighbors use at most five colors, leaving a color for v . $\square$

Toward Five Colors

The same induction immediately works if $\deg(v) \leq 4$.

If $\deg(v) = 5$ but two neighbors have the same color, it also works.

Question

What if the five neighbors use all five colors? Can we change part of the coloring?

Definition: A Two-Color Component

Two-color component

Take the subgraph induced by vertices with two chosen colors. A connected component of this subgraph is often called a Kempe component.

Interchanging the two colors throughout one component preserves properness.

Edges within the component still have different endpoint colors; edges leaving it cannot cause a conflict in those two colors.

Theorem: Five Colors Suffice

Five Color Theorem

Every simple planar graph is 5-colorable.

Induct on n . Delete a vertex v of degree at most five and color the rest.

Only the case of five neighbors with five distinct colors needs work. List these neighbors in cyclic order as $v_1, \dots, v_5$, with respective colors $1, \dots, 5$.

Proof: The First Color Swap

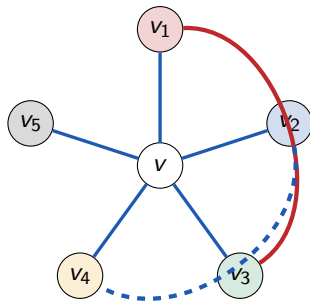
Proof

Consider colors 1 and 3. If v_1 and v_3 lie in different two-color components, swap colors 1 and 3 in the component containing v_1 .

No neighbor of v now has color 1. Give that color to v .

Otherwise, there is a path of colors 1, 3 joining v_1 to v_3 . We consider the second pair of colors.

Proof Continued: The Planar Separation



A simple 1–3 path, together with vv_1 and vv_3 , forms a closed curve.
The cyclic order puts v_2 and v_4 on opposite sides of it.

Proof Continued: The Second Color Swap

Proof

A path of colors 2, 4 from v_2 to v_4 would have to cross the closed curve. It cannot share a vertex with its 1, 3 path, and it cannot pass through the deleted vertex v . Planarity therefore forbids such a path.

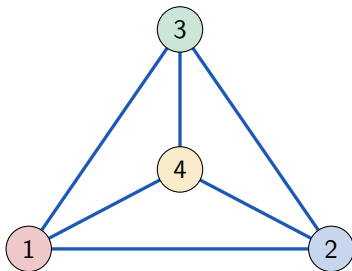
Swap colors 2 and 4 in the component containing v_2 . Color 2 becomes available for v . The induction is complete. □

The Four Color Theorem

Four Color Theorem

Every simple planar graph is 4-colorable.

This is considerably harder than the Five Color Theorem; we will use it as a theorem without proving it.



K_4 shows that four colors are sometimes necessary.

Definition: Chromatic Polynomial

Chromatic polynomial

For a positive integer k , let $P_G(k)$ be the number of proper colorings of G using the labeled colors $1, \dots, k$.

All k colors need not be used.

We will see that this counting function is a polynomial in k .

First Chromatic Polynomials

- For an edgeless graph on n vertices: $P_G(k) = k^n$.
- For K_n : $P_{K_n}(k) = k(k-1) \cdots (k-n+1)$.
- For a path on $n \geq 1$ vertices: $P_{P_n}(k) = k(k-1)^{n-1}$.

Color a first vertex, then make the remaining choices in sequence.

The Chromatic Polynomial of a Tree

Tree formula

For every tree T with n vertices,

$$P_T(k) = k(k-1)^{n-1}.$$

Proof

Root the tree and order parents before children. The root has k choices. Each other vertex need only avoid its parent's color, leaving $k-1$ choices. Unique root paths ensure that there are no other earlier neighbors. □

Deletion and Contraction

Deletion $G - e$

Remove an edge $e = uv$.

Contraction G/e

Identify u and v into a single vertex. Remove duplicate edges if they arise.

A coloring of $G - e$ either gives u, v different colors or gives them the same color.

Theorem: Deletion-Contraction

Deletion-contraction

For any edge e of a simple graph,

$$P_G(k) = P_{G-e}(k) - P_{G/e}(k).$$

Proof

Colorings of $G - e$ with different colors at the endpoints of e are exactly the colorings of G . Those with equal endpoint colors correspond bijectively to colorings of G/e . These two classes partition the colorings of $G - e$. □

Example: A Triangle

Delete one edge of K_3 . The result is P_3 ; contraction produces K_2 .

$$\begin{aligned}P_{K_3}(k) &= k(k-1)^2 - k(k-1) \\ &= k(k-1)(k-2).\end{aligned}$$

This agrees with assigning three different colors to three mutually adjacent vertices.

Example: A Four-Cycle

Deleting an edge of C_4 gives P_4 ; contracting it gives K_3 .

$$\begin{aligned}P_{C_4}(k) &= k(k-1)^3 - k(k-1)(k-2) \\ &= k(k-1)(k^2 - 3k + 3).\end{aligned}$$

In particular, $P_{C_4}(2) = 2$ and $P_{C_4}(3) = 18$.

The Chromatic Polynomial of a Cycle

Cycle formula

For $n \geq 3$,

$$P_{C_n}(k) = (k-1)^n + (-1)^n(k-1).$$

Proof

The formula holds for $n = 3$. For $n \geq 4$, deletion-contraction gives

$$P_{C_n}(k) = k(k-1)^{n-1} - P_{C_{n-1}}(k).$$

Insert the induction hypothesis and simplify.



Why It Is a Polynomial

Proof

Induct using deletion–contraction. An edgeless graph has polynomial k^n . Deletion removes an edge, while contraction decreases the number of vertices. Thus the recurrence eventually reaches edgeless graphs.

At each stage we subtract two polynomials, so $P_G(k)$ is a polynomial. Its leading term is k^n . $\square$

Information Encoded by the Polynomial

For a simple graph of order n and size m ,

$$P_G(k) = k^n - mk^{n-1} + \text{terms of lower degree.}$$

The chromatic number is the smallest positive integer k for which $P_G(k) > 0$.

Different graphs can have the same chromatic polynomial: all trees on n vertices do.

Practice: Chromatic Polynomials

Question

A graph consists of a triangle with a leaf attached to one vertex. Find its chromatic polynomial and chromatic number.

First color the triangle. Then color the leaf.

$$P_G(k) = k(k-1)(k-2)(k-1) = k(k-1)^2(k-2).$$

The triangle needs three colors and the leaf needs no additional color, so $\chi(G) = 3$.

This combines a local structural observation with the product rule.

Summary: Colorings

- Color classes are independent sets.
- Cliques and independence number give lower bounds; greedy coloring gives upper bounds.
- Planarity implies five colors by a short recoloring proof, and four by a deeper theorem.
- Chromatic polynomials count labeled proper colorings; deletion–contraction computes them.

Matchings